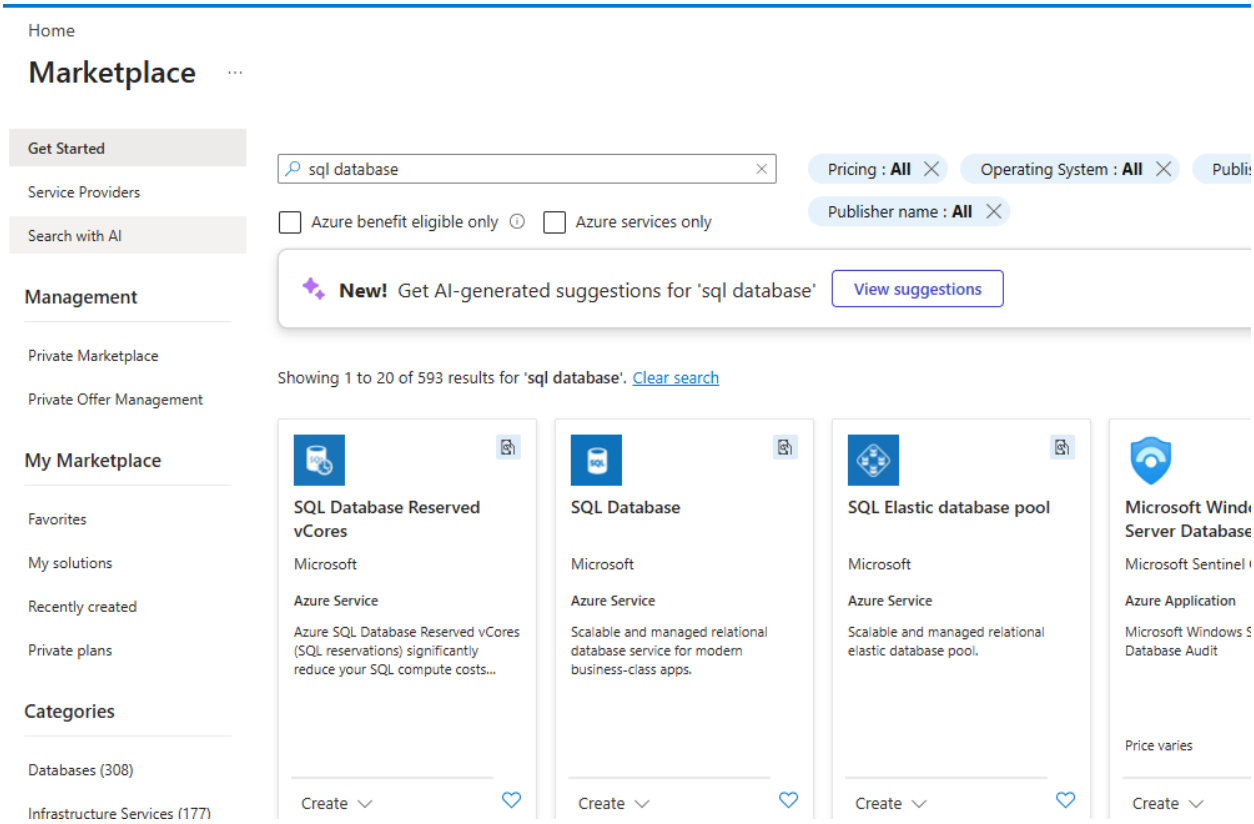
Name: Chauhan Sumit K.

Dept: Data Engineering

Task:

Creation of sql database in azure



It will prompt you with creation of server

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name *: sq-db-server-demo01

.database.windows.net

Location *: (Asia Pacific) Central India

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. Learn more

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication Learn more using an existing Microsoft Entra user, group, or application as Microsoft Entra admin Learn more

Authentication method

- (●) Use Microsoft Entra-only authentication
- ( ) Use both SQL and Microsoft Entra authentication
- ( ) Use SQL authentication

Set Microsoft Entra admin

Chauhansk1712_gmail.com#EXT#@Chauhansk1712gmail.onmicrosoft.com

Admin Object/App ID: 6c49096f-1fff-4cf3-8dea-7f909cdb35a3

Set admin

Create SQL Database

Microsoft

Basics | Networking | Security | Additional settings | Tags | Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. Learn more

Create a serverless Azure SQL Database with 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of your subscription. Limit is 10 free databases per subscription. Learn more

Apply offer

SQL Database Hyperscale: Low price, high scalability, and best feature set. Learn more

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure for Students

Resource group * : Select a resource group

Create new

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name * : az-sql-db

Server * : (new) az-sql-server-demo01 (Central India)

Create new

Cost summary

Hyperscale (HS_Gen5_2) - *Primary replica*

Cost per **vCore** (in USD)	146.68
vCores selected	x 2

Hyperscale (HS_Gen5_2) - *HA replicas*

Cost per **vCore** (in USD)	146.68
vCores selected	x 2
HA replicas	x 0
ESTIMATED COMPUTE COST / MONTH	293.35 USD
STORAGE COST / GB / MONTH	0.28 USD

STORAGE NOTES

[1] In the Hyperscale tier, storage costs are calculated based on actual allocation. Allocated space increases automatically as needed, up to 100 TB.

Compute + storage *

Basic

2 GB storage

Configure database

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy

- ○ Locally-redundant backup storage
- ○ Zone-redundant backup storage
- ◉ Geo-redundant backup storage
- ○ Geo-Zone-redundant backup storage

Selected value for backup storage redundancy is Geo-redundant backup storage. Database backups will be geo-replicated which might impact your data residency requirements. Learn more

Selecting the pricing model (here we have choose serverless)

Configure

Feedback

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. Learn more

SQL Database Hyperscale: Low price, high scalability, and best feature set. Learn more

Service tier: Basic (For less demanding workloads)

Compare service tiers

DTUs Compare DTU options

5 (Basic)

Data max size (GB): 2

Cost summary

Basic (Basic)

Cost per DTU (in USD)	1.12
DTUs selected	x 5
ESTIMATED COST / MONTH	5.60 USD

Review and create

Create SQL Database

Microsoft

Basics | Networking | Security | Additional settings | Tags | Review + create

Product details

SQL database
by Microsoft
Terms of use | Privacy policy

Estimated cost per month
5.60 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see Azure Marketplace Terms.

Basics

Subscription	Azure for Students
Resource group	rsg-sql-db
Region	Central India
Database name	az-sql-db
Server	(new) az-sql-server-demo01
Authentication method	Microsoft Entra-only authentication
Microsoft Entra Admin	sqladmin@Chauhansk1712gmail.onmicrosoft.com
Compute + storage	Basic: 2 GB storage
Backup storage redundancy	Geo-redundant backup storage

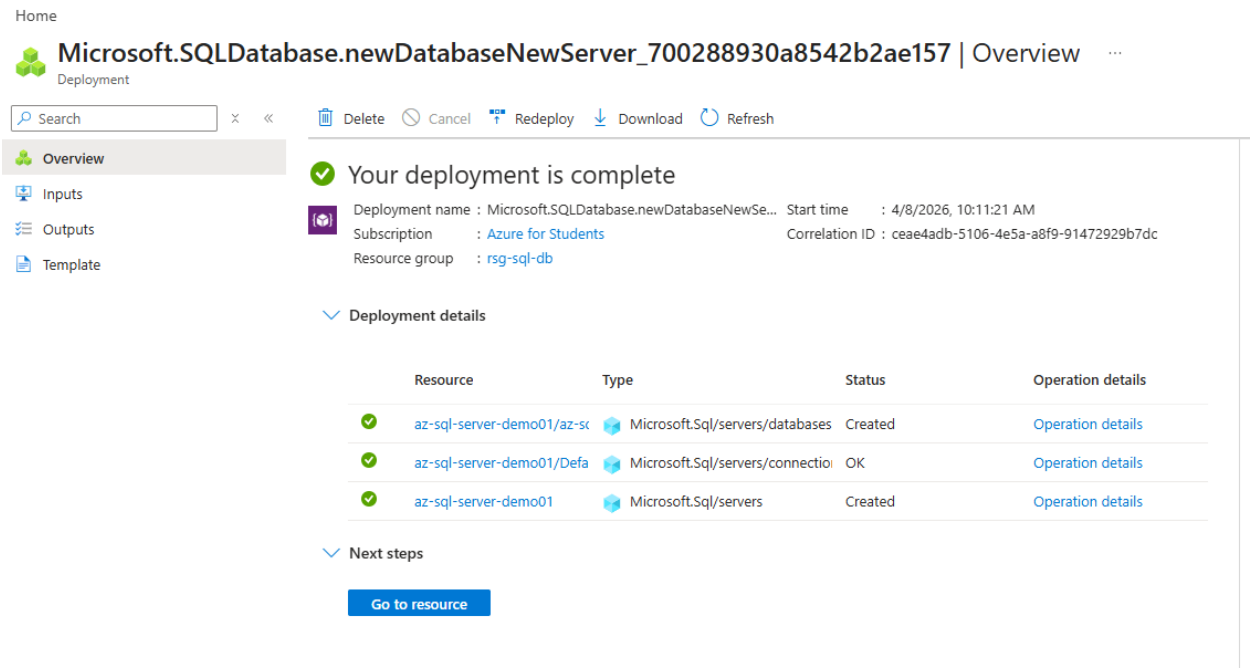
Networking

Cost summary

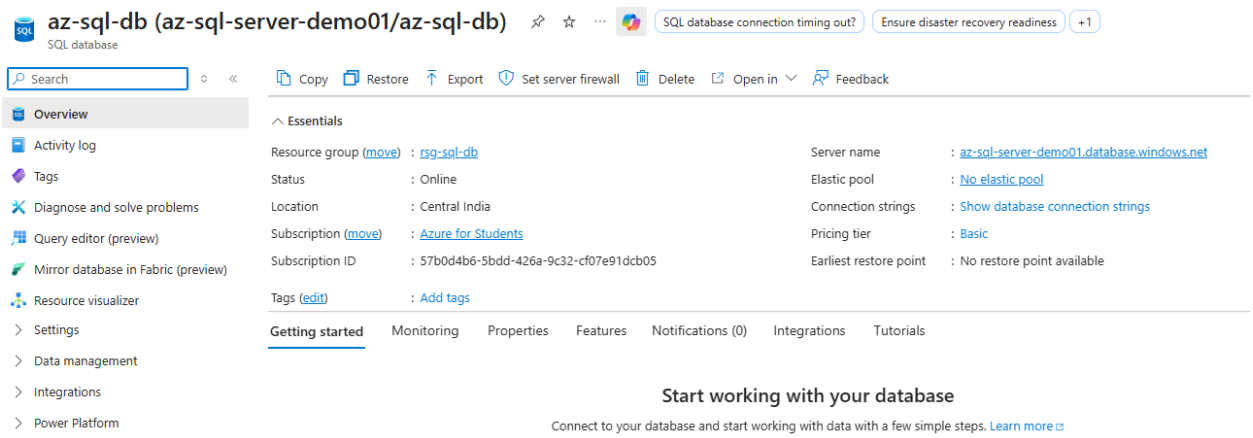
Basic (Basic)

Cost per DTU (in USD)	1.12
DTUs selected	x 5
ESTIMATED COST / MONTH	5.60 USD

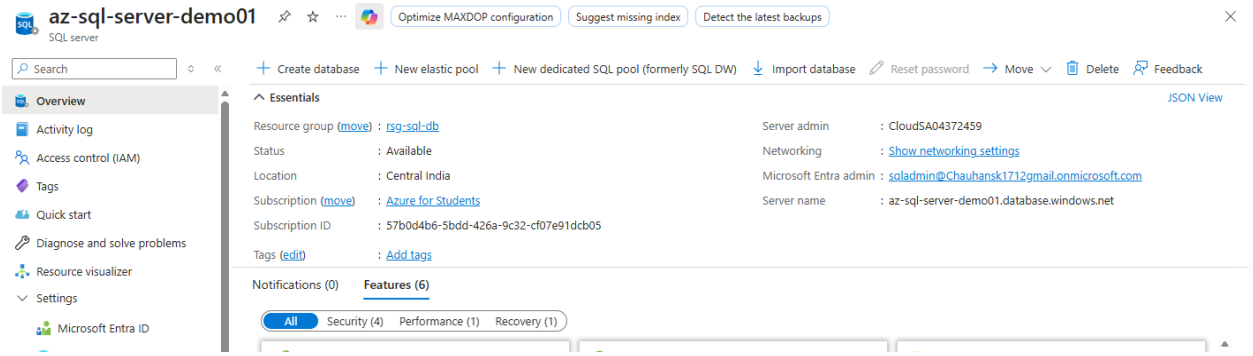
After deployment:



Sql-db:



Sql-server:

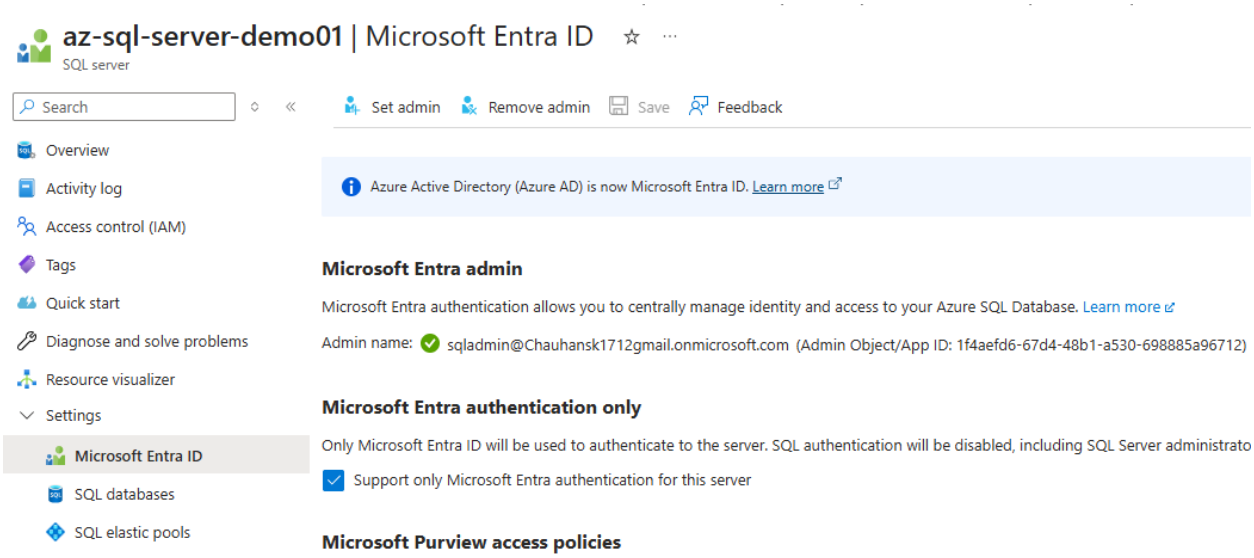


Connecting this Database to sql server (SSMS):

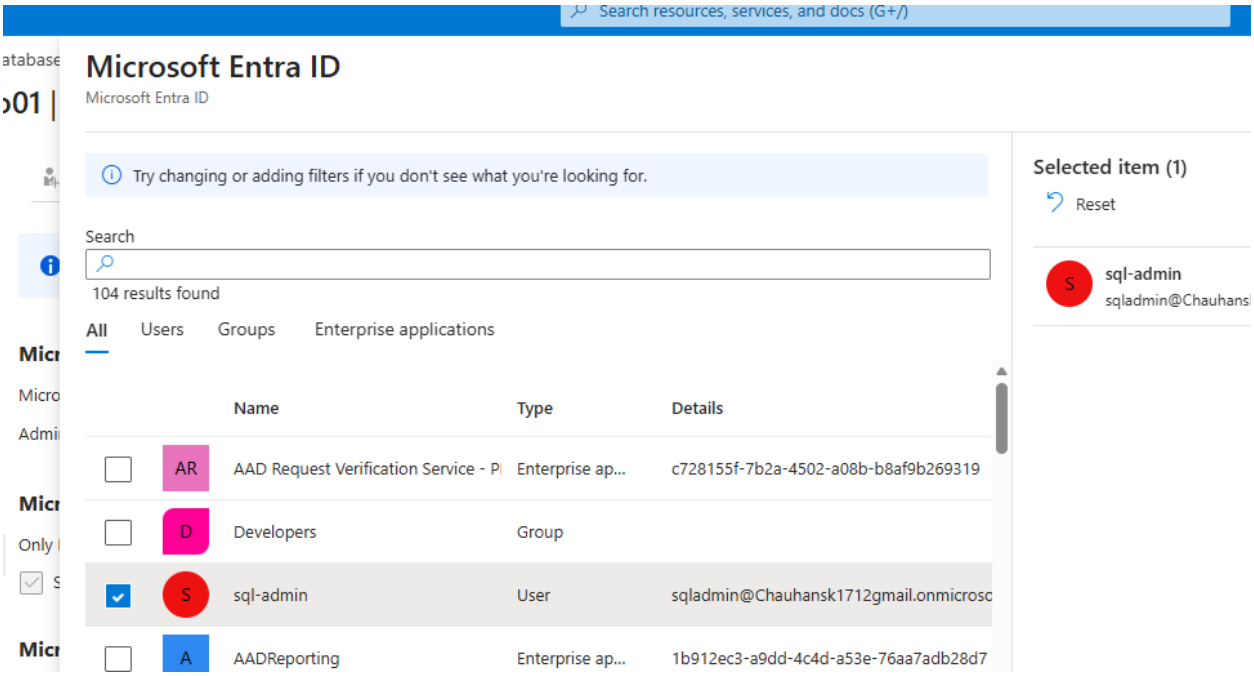
Firstly we need to setup an admin to our server

Go to your db-server

Left side bar > settings > microsoft entra ID



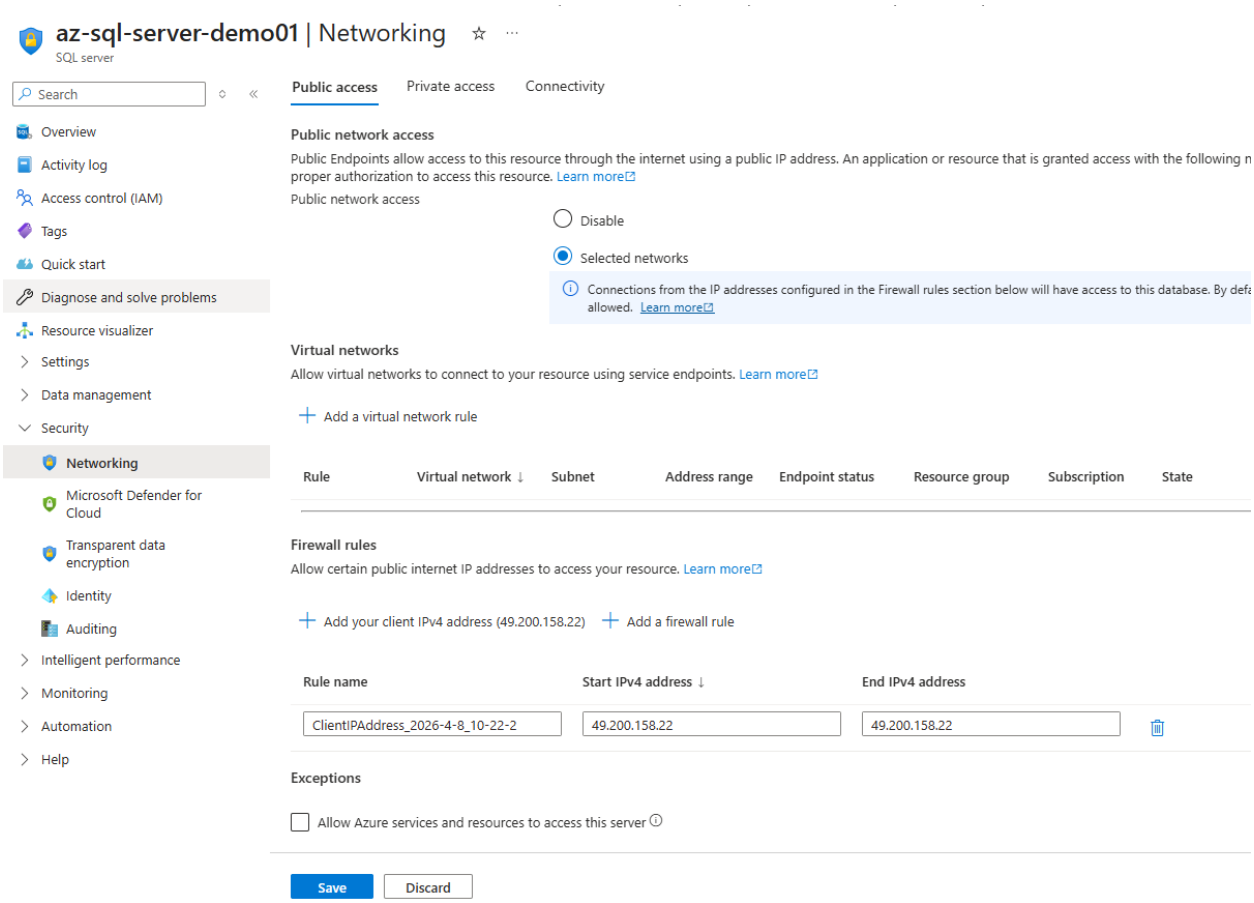
Click on set admin



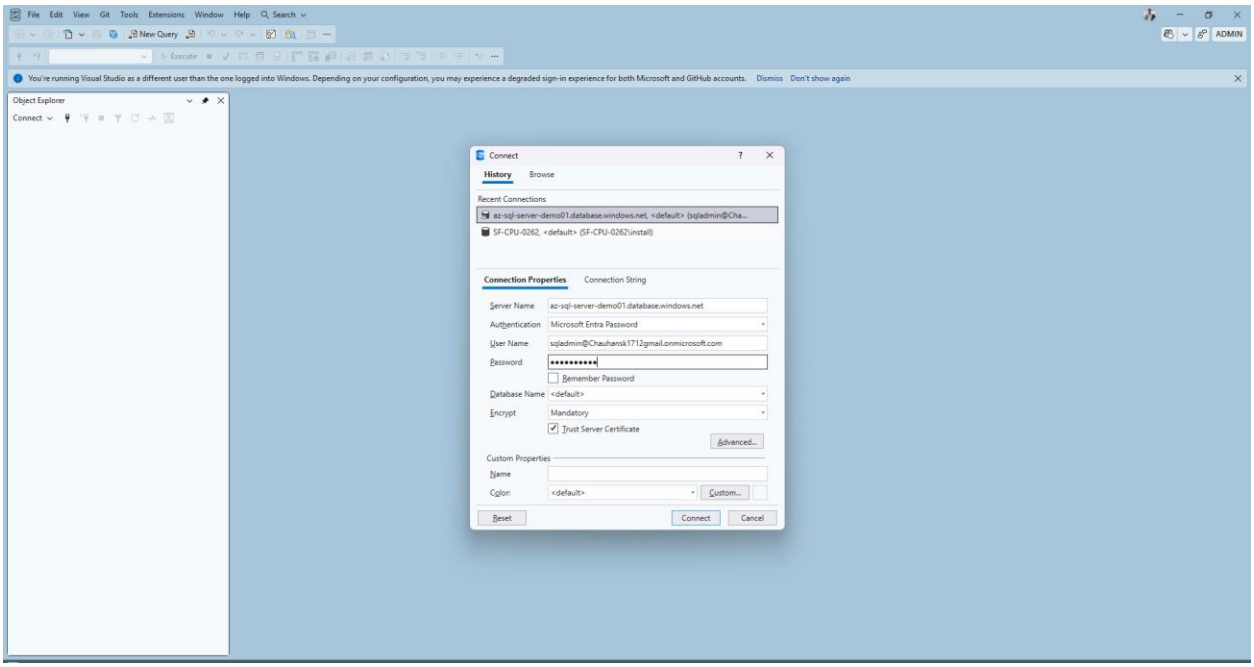
Hit on save!

Also we need to allow public access of this server

For that



Open SSMS on your machine:



Enter server name:

You can check the server name by opening your database in azure overview

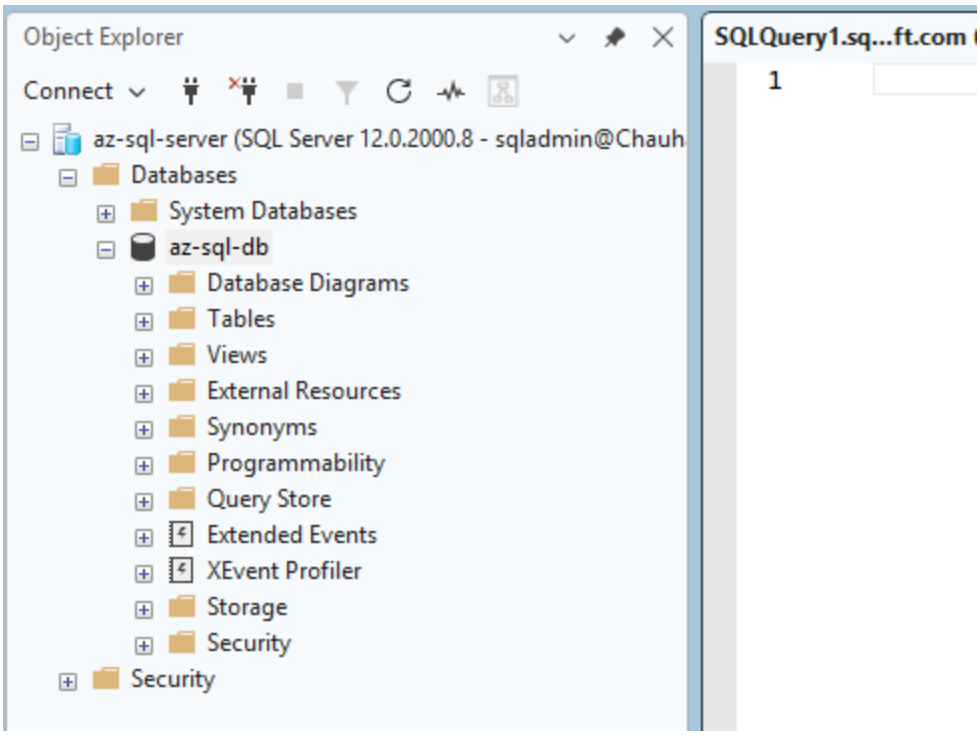
Server name	: az-sql-server-demo01.database.windows.net
Elastic pool	: No elastic pool
Connection strings	: Show database connection strings
Pricing tier	: Basic
Earliest restore point	: No restore point available

Select Microsoft entra ID password as authentication method.

MFA might not work

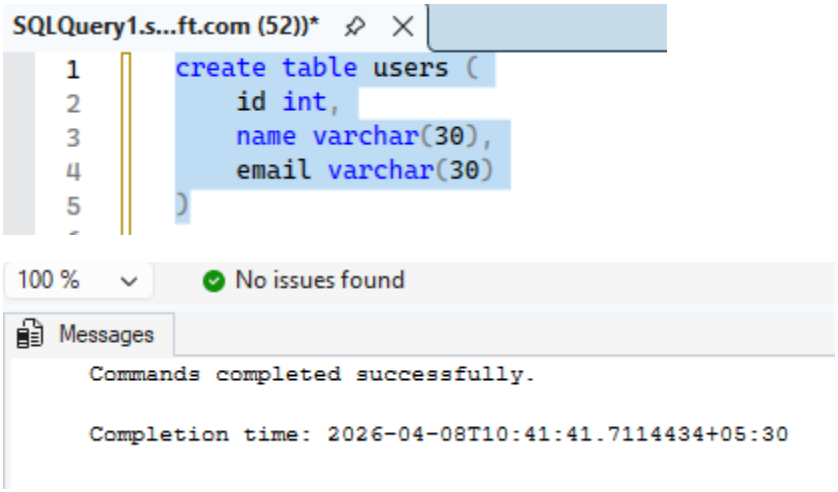
And enter the username and password of the server admin you configured

Server is connected with SSMS



Performing operations on database

Create:



Insert:

```sql
insert into users
values
    (1,'sumit','s@gmail.com'),
    (2,'mihir','m@gmail.com')
```

100 % No issues found

Messages

```
(2 rows affected)

Completion time: 2026-04-08T10:42:36.6869852+05:30
```

Select:

```sql
select * from users
```

100 % No issues found

Results Messages

	id	name	email
1	1	sumit	s@gmail.com
2	2	mihir	m@gmail.com

Update and select:

```sql
update users
set name = 'tushar', email = 't@gmail.com'
where name = 'mihir'
```

100 % No issues found

Results Messages

	id	name	email
1	1	sumit	s@gmail.com
2	2	tushar	t@gmail.com

Delete and select:

```
delete from users
where name = 'tushar'
```

100 % No issues found

Results | Messages

	id	name	email
1	1	sumit	s@gmail.com

Drop:

```
drop table users
```

100 % No issues found

Messages

```
Commands completed successfully.

Completion time: 2026-04-08T10:45:32.5376486+05:30
```

Creating an index on users table:

```
create index users_id_index
on users(id)

drop index users_id_index on users
```

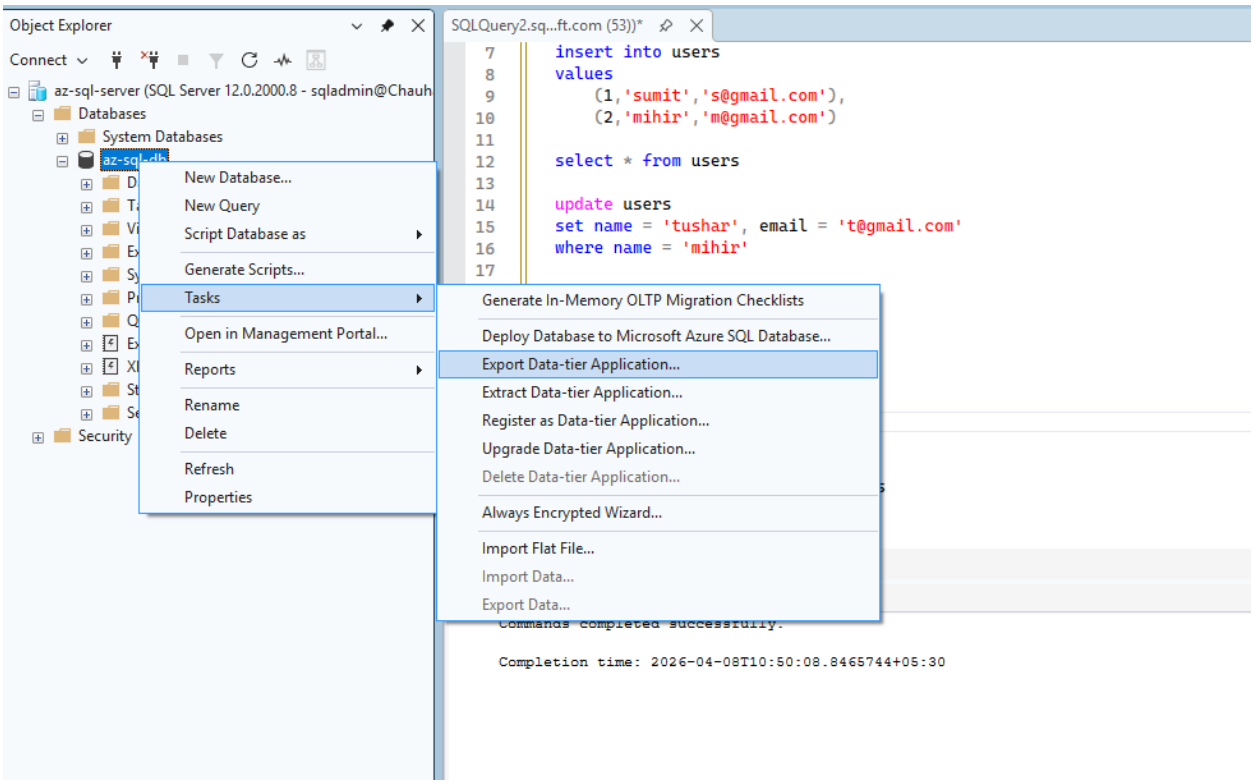
100 % 2 0

Messages

```
Commands completed successfully.

Completion time: 2026-04-08T10:50:08.8465744+05:30
```

Use SQL Server Management Studio (SSMS) to export data from your database to a .bacpac file.



Export Data-tier Application 'az-sql-db'

Export Settings

Introduction

Export Settings

Summary

Results

Help

Export Settings

This operation will create a BACPAC file that contains the logical contents of your database. To continue, specify the location where you want the BACPAC file to be created, and then click Next. To specify a subset of tables to export, use the Advanced option.

Settings | Advanced

- (●) Save to local disk

  C:\Users\sumit.chauhan\Desktop\LMS-modules\My-Courses\Azure-Data-Enginee [Browse...]

- ( ) Save to Microsoft Azure

  Storage account: [Connect...]

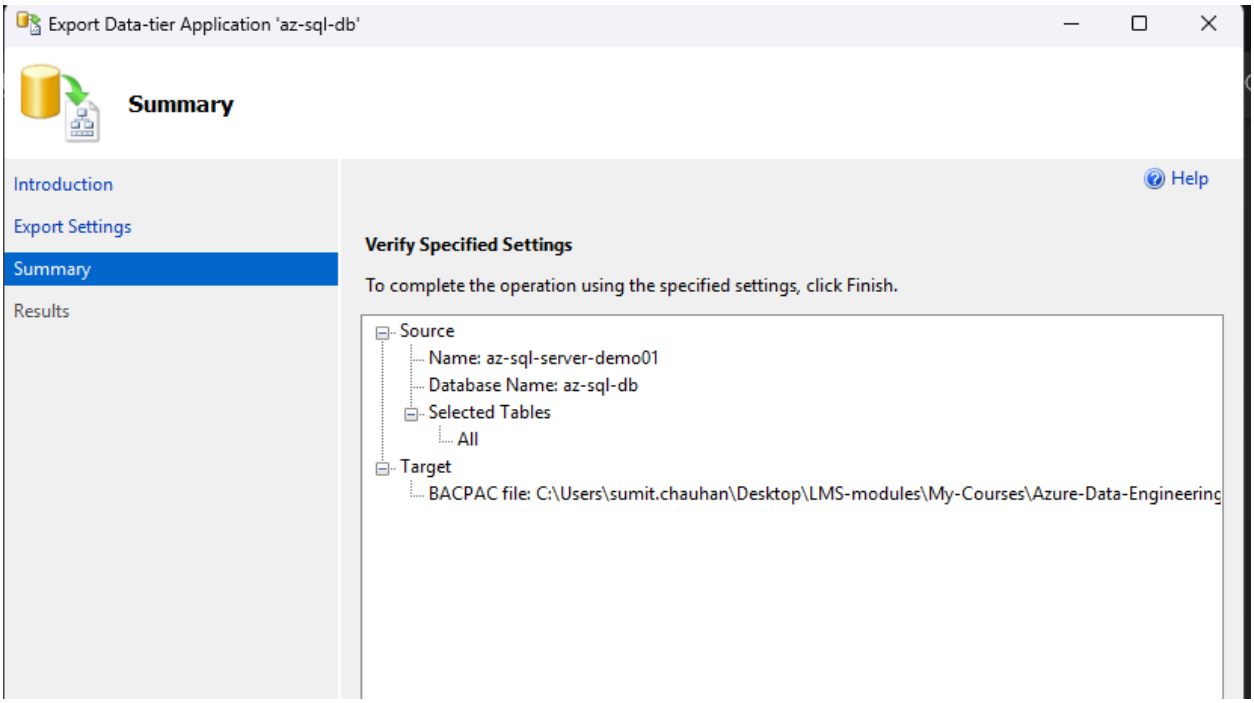
  Container:

  File name: az-sql-db.bacpac

  Temporary file name:

  C:\Users\install\AppData\Local\Temp\az-sql-db-20260408112118.bacpac [Browse...]

< Previous | Next > | Cancel


Export Data-tier Application 'az-sql-db'
Summary
Introduction
Export Settings
Summary
Results
Help
Verify Specified Settings
To complete the operation using the specified settings, click Finish.
Source
Name: az-sql-server-demo01
Database Name: az-sql-db
Selected Tables
All
Target
BACPAC file: C:\Users\sumit.chauhan\Desktop\LMS-modules\My-Courses\Azure-Data-Engineering

Export Data-tier Application 'az-sql-db'

Results

- Introduction
- Export Settings
- Summary
- Results

Help

Operation Complete

Summary:

Name	Result
Exporting database	Success
Extracting schema	Success
Extracting schema from database	Success
Resolving references in schema model	Success
Validating schema model	Success
Validating schema model for data package	Success
Validating schema	Success
Processing tables for data extraction	Success
Exporting data from database	Success
Exporting data	Success
Processing Export.	Success
Processing Table '[dbo].[users]'.	Success

< Previous | Next > | Close

Desktop > LMS-modules > My-Courses > Azure-Data-Engineering > Azure-SQL-database

Sort | View | ...

Name	Date modified	Type	Size
az-sql-db.bacpac	08-04-2026 11:28	BACPAC File	4 KB